

Final Project Progress Report

Analyzing U.S. County-Level Health Outcomes: The Role of Lifestyle, Environment, and Access to Care

The project scope remains consistent with the original proposal. The goal is to analyze relationships between county-level health outcomes, specifically premature death rates and diabetes prevalence, and factors such as physical inactivity, uninsured rates, air quality, and access to primary care across U.S. counties. No major changes have been made to the research questions or planned methods. The plan to use correlation analysis, multiple linear regression, and K-Means clustering remain in place.

Data Sources

County Health Rankings (File / CSV): Successfully downloaded and loaded the 2024 County Health Rankings dataset from countyhealthrankings.org. The dataset contains ~3,200 rows covering all U.S. counties with fields including premature death rate, diabetes prevalence, physical inactivity rate, uninsured rate, and primary care physician ratio. Data was cleaned by removing state-level summary rows and handling missing values.

CDC PLACES API (API): Successfully connected to the CDC PLACES API (data.cdc.gov/api) using Python's requests library. Retrieved county-level chronic disease measures including obesity rates, smoking rates, and annual checkup rates in JSON format. The API required pagination to collect the full dataset across all counties. Data was flattened and merged with the County Health Rankings dataset on county FIPS codes.

EPA Air Quality Annual Summary (File / CSV): Downloaded the EPA AQS annual summary file for 2023 from [aq5.aqmr.epa.gov](https://aq5.aqmr.epa.gov/aqs). Contains median AQI, max AQI, and counts of good/unhealthy days per county. Joined to the main dataset by county FIPS code, yielding coverage for ~1,000 counties with air quality monitoring stations.

Issues / Difficulties

The main challenge so far has been merging the three datasets, as each uses slightly different county identifier formats (some use FIPS codes as integers, others as zero-padded strings). This was resolved by standardizing all FIPS codes to 5-character zero-padded strings before joining. A secondary issue is that EPA air quality data only covers counties with active monitoring stations, leaving roughly two-thirds of counties without AQI values. The plan is to treat air quality as an optional feature and run analyses both with and without it. No issues were encountered accessing the CDC PLACES API.